波粒二象性　　电子是波+粒子.

波函数　　描述粒子波的函数.　　$\psi = Ae^{i(\vec{r}\cdot\vec{p} - Et)/\hbar}$.

波函数在空间中某点的强度与该点找到该粒子的概率成正比.

波函数性质:

1. 统计的结果.
2. 归一化
3. 波函数 ⟶ 体系右种性质.
4. 不同于经典力学描述方式，不能使用坐标与动量描写粒子，坐标与动量只是变量.
5. 波函数强度 $\propto$ 粒子出现概率　　概率波
6. 振幅同时加大，粒子出现概率不变

有意义的是相对强度

波函数性质的数学表示.

波函数在空间中某一点振幅的平方与该点找到粒子的概率成正比.

$$d\vec{F}(\vec{r},t) \propto |\phi(\vec{r},t)|^2 d\tau$$

$$d\vec{F}(\vec{r},t) = C|\phi(\vec{r},t)|^2 d\tau$$

$$w(\vec{r},t) = \frac{dW(\vec{r},t)}{d\tau} = C|\phi(\vec{r},t)|^2$$

由归一化条件:

$$\int_{-\infty}^{+\infty} dW(\vec{r},t) = \int_{-\infty}^{+\infty} C|\phi(\vec{r},t)|^2 d\tau = \int_{-\infty}^{+\infty} |\underline{\Psi}(\vec{r},t)|^2 d\tau = 1$$

归一化波函数.

$$\Longrightarrow \quad C = \frac{1}{\int_{-\infty}^{+\infty} |\psi(\vec{r},t)|^2 d\tau}$$

相因子.

$$|e^{i\delta}\Psi|^2 = |e^{i\delta}|^2|\Psi|^2 = |\Psi|^2 \quad (|e^{i\delta}|=1)$$

波函数乘上一个 $e^{i\delta}$ 物理意义不变, 归一化波函数可含任意相因子.

概率密度的相对性.

在某些情形,

$$\int_\Omega |\psi|^2 d\tau \longrightarrow \infty. \qquad C = 0.$$

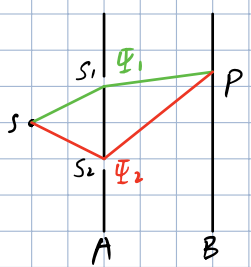
此时波函数无法归一化。但 $|\psi|^2$ 仍与概率成比例. 称相对概率密度.

态叠加原理

若 Ψ_1、Ψ_2 是粒子可能的状态，那么 $\Psi = c_1\Psi_1 + c_2\Psi_2$ 也是粒子可能的状态

粒子既在1态，又在2态，既不是1态，也不是2态.

例1：双狭缝实验.



$$\omega = |\Psi|^2 = |c_1\Psi + c_2\Psi|^2 = (c_1\Psi_1 + c_2\Psi_2)(c_1\Psi_1 + c_2\Psi_2)$$

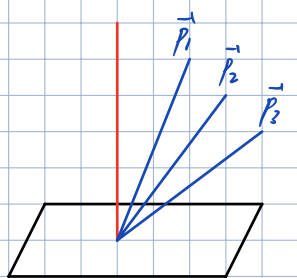
$$= (c_1\Psi_1 + c_2\Psi_2)^*(c_1\Psi_1 + c_2\Psi_2).$$

$$= (c_1^*\Psi_1^* + c_2^*\Psi_2^*)(c_1\Psi_1 + c_2\Psi_2)$$

$$= \underbrace{|c_1\Psi_1|^2}_{\text{粒子穿过}S_1\text{出现在}P} + \underbrace{|c_2\Psi_2|^2}_{\text{粒子穿过}S_2\text{出现在}P} + \underbrace{c_1^*c_2\Psi_1^*\Psi_2 + c_1c_2^*\Psi_1\Psi_2^*}_{\text{干涉项}}$$

推论：$\Psi = \sum_n c_n\Psi_n$

例2：晶体表面衍射.



反射后的粒子有各种动量.(即状态，用 $\vec{p}$ 表示).

$$\Psi_{\vec{p}}(\vec{r}, t) = e^{i(Et - \vec{p}\cdot\vec{r})/\hbar}.$$

叠加的态形成衍射条纹.

$$\Psi_p(\vec{r}, t) = \sum_{\vec{p}} c(\vec{p})\,\Psi_{\vec{p}}(\vec{r}, t)$$

证明：Ψ - 波函数 $\Psi(r,t)$ 可看作各种不同动量的平面波叠加.

薛定谔方程 Schrodinger equation

对薛定谔方程的猜测.

① 动量.

$$\frac{\partial \Psi}{\partial x} = \frac{i}{\hbar} p_x \Psi \quad ; \quad \frac{\partial \Psi}{\partial y} = \frac{i}{\hbar} p_y \Psi \quad . \quad \frac{\partial \Psi}{\partial z} = \frac{i}{\hbar} p_z \Psi .$$

$$\Rightarrow \qquad \hat{p}\Psi = -i\hbar \left(\hat{e}_x \frac{\partial}{\partial x} + \hat{e}_y \frac{\partial}{\partial y} + \hat{e}_z \frac{\partial}{\partial z} \right) \Psi = -i\hbar \left(\nabla \cdot \Psi \right)$$

② 动量的平方.

$$\frac{\partial^2 \Psi}{\partial x^2} = -\frac{1}{\hbar^2} p_x^2 \Psi . \qquad \frac{\partial^2 \Psi}{\partial y^2} = -\frac{1}{\hbar^2} p_y^2 \Psi . \qquad \frac{\partial^2 \Psi}{\partial z^2} = -\frac{1}{\hbar^2} p_z^2 \Psi .$$

$$\hat{p^2}\Psi = -\hbar^2 \left(\nabla \cdot \nabla \Psi \right) = -\hbar^2 \nabla^2 \Psi .$$

③ 能量.

$$\frac{\partial \Psi}{\partial t} = -\frac{i}{\hbar} E \Psi .$$

$$\Rightarrow \qquad E\Psi = i\hbar \frac{\partial \Psi}{\partial t} .$$

自由粒子. $$E = \frac{p^2}{2m_\mu} + U(r)$$

$\Rightarrow$

薛定谔波动方程:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m_\mu} \nabla^2 \Psi + U(r) \Psi$$

多粒子.

$$i\hbar\frac{\partial\Psi}{\partial t} = -\sum_{i=1}^{N}\frac{\hbar^2}{2m_i}\nabla_i^2\Psi + U(r_1, r_2, \cdots, r_N)\Psi.$$

∇_i 指对第 i 个粒子的拉普拉斯算符.

波函数描述粒子

概率随时间变化.

概率流密度，连续性方程.

$$\frac{\partial w}{\partial t}=\frac{\partial(|\Psi|)^2}{\partial t}=\frac{\partial(\Psi^*\Psi)}{\partial t}=\Psi^*\frac{\partial\Psi}{\partial t}+\Psi\frac{\partial\Psi^*}{\partial t}$$

Schrodinger equation：

$$\begin{cases}\dfrac{\partial\Psi}{\partial t}=-\dfrac{i\hbar}{2m}\nabla^2\Psi+\dfrac{1}{i\hbar}U(r)\Psi \\ \dfrac{\partial\Psi^*}{\partial t}=-\dfrac{i\hbar}{2m}\nabla^2\Psi^*+\dfrac{1}{i\hbar}U(r)\Psi^*\end{cases}$$

代入，有：

$$\frac{\partial w}{\partial t}=\frac{i\hbar}{2m}(\Psi^*\nabla^2\Psi-\Psi\nabla^2\Psi^*)=\frac{i\hbar}{2m}\nabla\cdot(\Psi^*\nabla\Psi-\Psi\nabla\Psi^*)$$

设概率流密度矢量 J

$$J\equiv\frac{i\hbar}{2m}(\Psi\nabla\Psi^*-\Psi^*\nabla\Psi)$$

有连续性方程

$$\frac{\partial w}{\partial t}+\nabla\cdot J=0.$$

对空间中任意体积 V 积分.

$$\int_V\frac{\partial w}{\partial t}d\tau=\frac{\partial}{\partial t}\int_V w\,d\tau=-\int_V\nabla\cdot J\,d\tau.$$

应用 Gauss's law.

$$\int_V \frac{\partial w}{\partial t} d\tau = -\oint_S J \cdot dS = -\oint_S J_n dS.$$

dS 为包围体积 dV 的封闭曲面面元.

由上我们意识到:

J 的意义为概率流密度矢量，其在 S 面上法向分量面积分表单位时间内流过 S 面上单位面积的概率.

若 $V \to \infty$.

$$\frac{d}{dt}\int_\infty w d\tau = -\frac{d}{dt}\int_\infty \Psi^* \Psi \, d\tau = 0$$

即在整个空间中找到粒子的概率与时间无关.

波函数的归一性恒保持不变.

定义 质量流密度

$$J_m \equiv mJ = \frac{i\hbar}{2}(\Psi \nabla \Psi^* - \Psi^* \nabla \Psi)$$

$$\frac{\partial w_m}{\partial t} + \nabla \cdot J_m = 0 \qquad \text{(质量守恒定律)}$$

类似上面地，我们有: 单位时间内体积 V 内质量的增加等于穿过 V 的边界 S 流进的质量.

同样的.

定义 电荷密度

$$w_g = g w$$

电荷守恒定律.

$$\frac{\partial w_g}{\partial t} + \nabla \cdot J_g = 0.$$

波函数3个标准条件.

①有限性. ②连续性 ③单值性.

定态薛定谔方程

若 $U(r)$ 与时间无关.

分离变量，考虑

$$\Psi(r,t)=\psi(r)T(t).$$

$$\frac{i\hbar}{T}\frac{dT}{dt}=\frac{1}{\psi}\left[-\frac{\hbar^2}{2m}\nabla^2\psi+U(r)\psi\right]$$

左边只是 t 函数，右边只是 r 函数，左右恒相等.

定态：能量具确定值.

$$\begin{cases} i\hbar\frac{dT}{dt}\equiv ET. \\ -\frac{\hbar^2}{2m}\nabla^2\psi+U(r)\psi=E\psi \end{cases}$$

$$\frac{i\hbar}{T}\frac{dT}{dt}\equiv ET. \Longrightarrow T(t)=Ce^{-\frac{iE}{\hbar}t}.$$

$$\Psi(r,t)=C\psi(r)e^{-\frac{iE}{\hbar}t}$$

定态波函数满足：

$$i\hbar\frac{\partial\Psi}{\partial t}=E\Psi.$$

$$\left[-\frac{\hbar^2}{2m}\nabla^2+U(r)\right]\Psi=E\Psi.$$

定态：能量具有确定值的状态.

时间部份

$$E=\frac{i\hbar}{T}\frac{dT}{dt}=\frac{i\hbar\, d(\ln T)}{dt}$$

$$d\ln T=-\frac{iE}{\hbar}dt \qquad \ln T=-\frac{iE}{\hbar}t+A$$

$$T=Ce^{-\frac{iEt}{\hbar}}$$

$$\Psi=\psi(\vec{r})e^{-\frac{iEt}{\hbar}},$$

$$\omega=\frac{E}{\hbar}.$$

定态的概率密度与概率流密度与时间无关.

ψ 可由具体问题的边界条件得到.

E 由空间部份方程得到.

空间部份

$$E\psi=-\frac{\hbar}{2m}\nabla^2\psi+U\psi$$

$$\begin{cases} i\hbar\dfrac{\partial\Psi}{\partial t}=\psi i\hbar\dfrac{dT}{dt}=\psi E=E\Psi \\ -\dfrac{\hbar^2}{2m}\nabla^2\Psi+U\Psi=-\psi\dfrac{\hbar^2}{2\mu}\nabla^2\psi+\psi U=\psi E=E\Psi \end{cases}$$

定义哈密顿算符：

$$\hat{H} \equiv i\hbar\frac{\partial}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2 + U(r).$$

哈密顿算符作用于波函数Ψ上得到能量E，又称能量算符。

$$\hat{H}\Psi = E\Psi.$$

本征值方程

E称为$\hat{H}$的本征值，Ψ称为$\hat{H}$的本征函数。

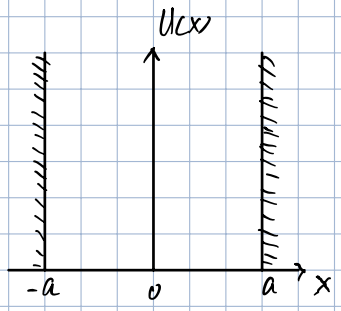
本征值一般有多个，第n个记为E_n，相应本征函数记Ψ_n.

$$\Psi_n(\vec{r},t) = \psi_n(\vec{r})e^{-i\frac{E_n t}{\hbar}}$$

含时薛定谔方程一般解：

$$\Psi = \sum_n c_n\Psi_n(\vec{r},t) = \sum_n c_n\Psi_n(\vec{r})e^{-i\frac{E_n t}{\hbar}}$$

一维无限深方势阱



$$U(x)=\begin{cases}0 & |x|<a\\ \infty & |x|\geq a\end{cases}$$

定态薛定谔方程：

$$E\psi=-\frac{\hbar^2}{2\mu}\nabla^2\psi+U\psi=-\frac{\hbar^2}{2\mu}\frac{d^2\psi}{dx^2}+U\psi$$

$$\Rightarrow\begin{cases}E\psi=-\frac{\hbar^2}{2\mu}\frac{d^2\psi}{dx^2} & |x|<a \quad ①\\ E\psi=-\frac{\hbar^2}{2\mu}\frac{d^2\psi}{dx^2}+\infty\psi & |x|\geq a \quad ②\end{cases}$$

②：$\psi=0$，对 $|x|\geq a$。（边界条件）。

求解①

(1) $E>0$ 情形，化为：

$$0=\frac{d\psi^2}{dx^2}+\frac{2\mu E}{\hbar^2}\psi=\frac{d\psi^2}{dx^2}+\beta^2\psi.$$

$$\beta=\sqrt{\frac{2\mu E}{\hbar^2}}$$

考虑解形式

$$\psi=A\sin\beta x+B\cos\beta x.$$

代入，有：

$$-\beta^2(A\sin\beta a+B\cos\beta a)+\beta^2(A\sin\beta a+B\cos\beta a)=0$$

等合方程.

现考虑边界条件： $\psi(\pm a)=0$ 有：

$$\begin{cases} A\cos\beta a + B\sin\beta a = 0 \\ A\cos\beta a - B\sin\beta a = 0 \end{cases} \Longrightarrow \begin{cases} A\cos\beta a = 0 \\ B\sin\beta a = 0 \end{cases}$$

① $A=B=0$. $\psi=0$ trivial solution

② $A\neq 0$, $B\neq 0$.

$$\beta a = \frac{2n+1}{2}\pi = m\pi$$

no solution

③ $A=0$ $B\neq 0$

$$\beta a = n\pi, \quad n=1,2,3,\cdots$$

④ $A\neq 0$ $B=0$

$$\beta a = \frac{2n+1}{2}\pi, \quad n=1,2,3,\cdots$$

合并：

$$\beta a = \frac{n}{2}\pi, \quad n=1,2,3,\cdots.$$

$$\beta = \frac{n\pi}{2a}, \quad n=1,2,3,\cdots$$

$$E_n = \frac{\hbar^2 n^2 \pi^2}{8\mu a^2}, \quad n=1,2,3,\cdots$$

代入边界条件有:

$$\psi_n = \begin{cases} A' \sin \frac{n\pi}{2a}(x+a), & n \in N, \ |x| < a \\ 0, & |x| \geq a \end{cases} \qquad (A' = \frac{1}{\sqrt{a}})$$

$$\Psi_n(x,t) = \psi_n(x) e^{-\frac{i}{\hbar}E_n t}$$

$$= \frac{1}{\sqrt{a}} \sin \frac{n\pi}{2a}(x+a) e^{-\frac{i}{\hbar}E_n t}$$

$$= \frac{1}{\sqrt{a}} \sin \frac{n\pi(x+a)}{2a} e^{-i\frac{\hbar n^2 \pi^2 t}{8\mu a^2}}$$

(2) 考虑 $E=0$ 情形

$$\frac{d^2\psi}{dx^2} = 0$$

$$\frac{d\psi}{dx} = C$$

$$\psi = \frac{C}{2}x^2$$

$$\Psi(x,t) = \frac{C}{2}x^2 e^{-iEt/\hbar} = \frac{C}{2}x^2$$

(3) 考虑 $E<0$ 情形

$$0=\frac{d\psi^2}{dx^2}+\frac{2\mu E}{\hbar^2}\psi$$

令 $\beta=\sqrt{-\frac{2\mu E}{\hbar^2}}$

$$\frac{d\psi^2}{dx^2}-\beta^2\psi=0$$

考虑解形式：

$$\psi=Ae^{kx}$$

$$e^{kx}(k^2-\beta^2)=0$$

$$k_1=\beta,\ k_2=-\beta.$$

$$\psi=Ae^{\beta x}+Be^{-\beta x}$$

边界条件. $\psi(\pm a)=0$. 代入，有：

$$\begin{cases} Ae^{\beta a}+Be^{-\beta a}=0 \\ Ae^{-\beta a}+Be^{\beta a}=0 \end{cases}$$

$$(A+B)(e^{\beta a}+e^{-\beta a})=0.$$

$$(A-B)(e^{\beta a}-e^{-\beta a})=0$$

$$A=B=0$$

trivial solution

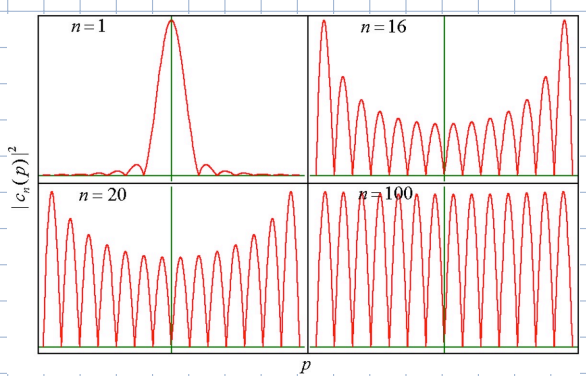
基本粒子动量分布

每种动量出现的概率？

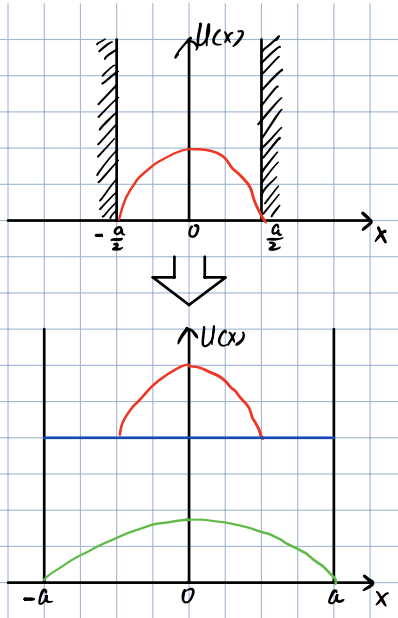
$$\psi_1(x) = \frac{1}{\sqrt{a}}\cos\left(\frac{\pi x}{2a}\right) \qquad \phi_2(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{-ipx}{\hbar}}$$

$$c_1(p) = \int_{-\infty}^{\infty} \psi_1(x)\,\phi_2(x)\,dx = \frac{1}{\sqrt{2\pi\hbar}} \int_{-a}^{a} \psi_1(x)\, e^{-ipx/\hbar}\,dx = \frac{1}{\sqrt{2\pi\hbar}} \int_{-a}^{a} \frac{1}{\sqrt{a}}\cos\left(\frac{\pi x}{2a}\right) e^{-ipx/\hbar}\,dx$$

$$= \frac{2\hbar\sqrt{2\pi a\hbar}}{-4a^2p^2+\pi^2\hbar^2}\cos\left(\frac{ap}{\hbar}\right)$$



例：一维初态演化问题.



一维单粒子.

势能

$$U(x)=\begin{cases}0 & |x|<\frac{a}{2}\\ \infty & else\end{cases}$$

粒子处于基态 ψ_1

势阱宽度突然增宽至 a. 求粒子留在新基态 ψ_1' 概率?

① 粒子状态来不及改变，波函数不变

用新系统的本征函数 ψ_1' 展开旧波函数 ψ_1，展开系数便是留在新基态的概率.

② 核心：态叠加原理

旧系统粒子基态 ψ_1,

新系统 ψ_2

$$\psi_1(x)=\begin{cases}\frac{1}{\sqrt{\frac{a}{2}}}\cos\frac{\pi x}{a}, & |x|<\frac{a}{2}\\ 0, & else\end{cases}\qquad \psi_1'(x)=\begin{cases}\frac{1}{\sqrt{a}}\cos\frac{\pi x}{2a}, & |x|<a\\ 0, & else\end{cases}$$

用新本征函数 ψ_1' 展开旧波函数 ψ_1

$$C_1=\int_{-a}^{a}\psi_1'(x)\,\psi_1(x)\,dx=\int_{-\frac{a}{2}}^{\frac{a}{2}}\frac{1}{\sqrt{a}}\cos\frac{\pi x}{2a}\,\frac{1}{\sqrt{\frac{a}{2}}}\cos\frac{\pi x}{a}\,dx=\frac{8}{3\pi}$$

$$=\int_{-\frac{a}{2}}^{\frac{a}{2}}\frac{1}{\sqrt{a}}\cos\left(\frac{\pi x}{2a}\right)\frac{1}{\sqrt{\frac{a}{2}}}\cos\left(\frac{\pi x}{a}\right)dx.$$

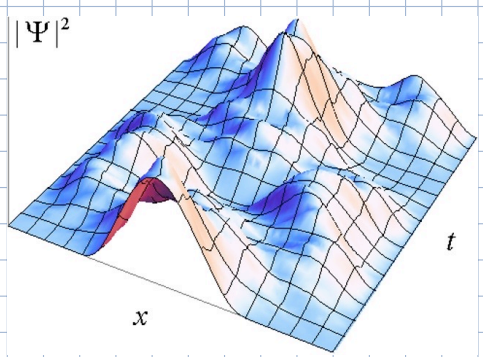
$$= \frac{\sqrt{2}}{a}\int_{-\frac{a}{2}}^{\frac{a}{2}} \cos\left(\frac{\pi x}{2a}\right)\cos\left(\frac{\pi x}{a}\right)dx = \frac{\sqrt{2}}{a}\,\frac{4\sqrt{2}}{3\pi}a = \frac{8}{3\pi}$$

$$|C_1|^2 = \frac{64}{9\pi^2} \approx 0.721$$

初态 ψ_1 的演化（含时间波函数）

$$\Psi(x,t) = \sum_n C_n \psi_n(x) e^{-iE_n t/\hbar}$$

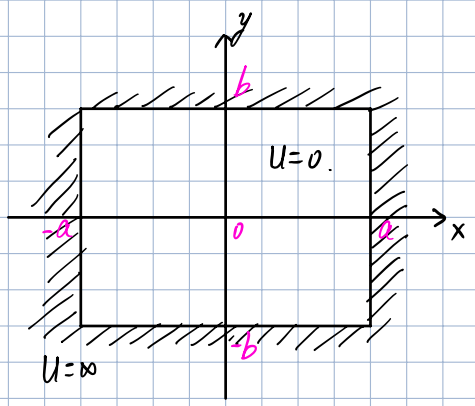
$$C_n = \int_{-a}^{a} \psi_n(x)\psi_1(x)\,dx = \frac{4\sqrt{2}\left(\sin\frac{n\pi}{4} + \sin\frac{3n\pi}{4}\right)}{\pi(n^2-4)}$$



例：二维定态问题.

二维单粒子

$$U(x)=\begin{cases}0 & |x|<a,\ |y|<b\\ \infty\end{cases}$$



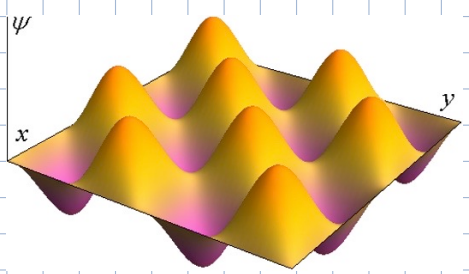
看作 2 - 维叠加.

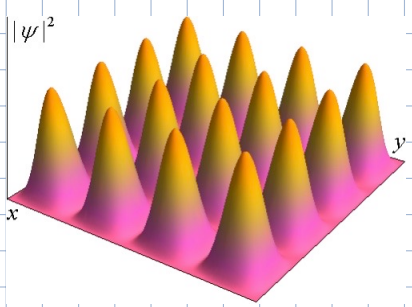
(对 x、y 分离变量)

本征值与本征函数.

$$E_{n_x,n_y}=\frac{\hbar^2\pi^2}{8\mu}\left(\frac{n_x^2}{a^2}+\frac{n_y^2}{b^2}\right),\quad n_x,n_y=1,2,3,\cdots.$$

$$\psi_{n_x,n_y}=\frac{1}{\sqrt{ab}}\sin\frac{n_x\pi(x+a)}{2a}\sin\frac{n_y\pi(y+b)}{2b}$$

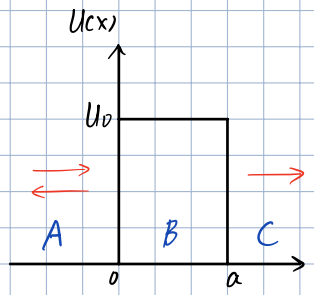




势垒贯穿

一维单粒子.

$$U(x)=\begin{cases}U_0 & 0<x<a\\ 0 & \text{else}\end{cases}$$



Schrodinger eq

$$\begin{cases}\dfrac{\hbar^2}{2m}\dfrac{d^2\psi}{dx^2}+E\psi=0 & x<0 \quad x>a\\[2ex] \dfrac{\hbar^2}{2m}\dfrac{d^2\psi}{dx^2}+(E-U_0)\psi=0 & 0<x<a\end{cases}$$

经典视角下，只有能量 $E>U_0$ 的粒子才能越过势垒进入 $x>a$ 区域

量子力学中，无论粒子能量 E 是否 $>U_0$，粒子均有可能贯穿势垒.

(1) $E>U_0$.

上式化为

$$\begin{cases}\dfrac{d^2\psi}{dx^2}+\dfrac{2mE}{\hbar^2}\psi=0 \Longrightarrow \dfrac{d^2\psi}{dx^2}+k_1^2\psi=0 & 0<x<a\\[2ex] \dfrac{d^2\psi}{dx^2}+\dfrac{2m}{\hbar}(E-U_0)\psi=0 \Longrightarrow \dfrac{d^2\psi}{dx^2}+k_2^2\psi=0 & x<0,\ x>a.\end{cases}$$

$x<0$ $\quad \psi_1 = Ae^{ik_1x} + A'e^{-ik_1x}$

$0<x<a$ $\quad \psi_2 = Be^{ik_2x} + B'e^{-ik_2x}$

$x>a$ $\quad \psi_3 = Ce^{ik_1x} + C'e^{-ik_1x}$

由左向右 由右向左.

对 ψ_1, Ae^{-ik_1x} 为入射波, $A'e^{-ik_1x}$ 为反射波.

而对 ψ_3, 由于只应有透射波, 不应用向左传播的波.

$$C' = 0.$$

粒子在某处概率密度是连续的, 故粒子波函数的微商也应当为连续的.

证明: 由 $\psi'' + k^2\psi = 0$, 有:

$$0 = \int_{x-\varepsilon}^{x+\varepsilon} (\psi'' + k^2\psi)\,d\xi = \int_{x-\varepsilon}^{x+\varepsilon} \psi''\,d\xi + k^2\int_{x-\varepsilon}^{x+\varepsilon} \psi\,d\xi$$

$$= \psi'\Big|_{x-\varepsilon}^{x+\varepsilon} + k^2\psi(x')2\varepsilon \qquad (x-\varepsilon < x' < x+\varepsilon) \quad ?$$

$$\psi'(x+\varepsilon) - \psi'(x-\varepsilon) = -2k^2\psi(x')\varepsilon.$$

取极限 $\varepsilon \to 0$.

$$\lim_{\varepsilon\to 0} \psi'(x+\varepsilon) = \psi'(x^+);$$

$$\lim_{\varepsilon\to 0} \psi'(x-\varepsilon) = \psi'(x^-);$$

$$\lim_{\varepsilon\to 0} \psi(x') = \psi(x)$$

于是，

$$\psi'(x^+) - \psi'(x^-) = -2\psi(x)\lim_{\varepsilon\to 0}(k^2\varepsilon)\begin{cases} = 0 & k^2 < \infty \\ \neq 0 & k^2 = \infty \end{cases}$$

只要 $k^2 < \infty$，波函数的微商 $\frac{d\psi}{dx}$ 便为连续的.

$$\begin{cases} (\psi_1)_{x=0} = (\psi_2)_{x=0} \Longrightarrow A + A' = B + B' \\ (\psi_2)_{x=a} = (\psi_3)_{x=a} \Longrightarrow k_2(Be^{ik_2a} - B'e^{-ik_2a}) = k_1 Ce^{ik_1a} \\ \left(\frac{d\psi_1}{dx}\right)_{x=0} = \left(\frac{d\psi_2}{dx}\right)_{x=0} \Longrightarrow k_1(A - A') = k_2(B - B') \\ \left(\frac{d\psi_2}{dx}\right)_{x=a} = \left(\frac{d\psi_3}{dx}\right)_{x=a} \Longrightarrow k_2Be^{ik_2a} - k_2B'e^{-ik_2a} = k_1Ce^{ik_1a} \end{cases}$$

解得：

$$A' = \frac{-i2(k_1^2 - k_2^2)\sin k_2a}{(k_1+k_2)^2e^{-ik_2a} - (k_1-k_2)^2e^{ik_2a}}A$$

$$B = \frac{2k_1(k_1+k_2)e^{-ik_2a}}{(k_1+k_2)^2e^{-ik_2a} - (k_1-k_2)^2e^{ik_2a}}$$

$$B' = \frac{-2k_1(k_1-k_2)e^{-ik_2a}}{(k_1+k_2)^2e^{-ik_2a} - (k_1-k_2)^2e^{ik_2a}}$$

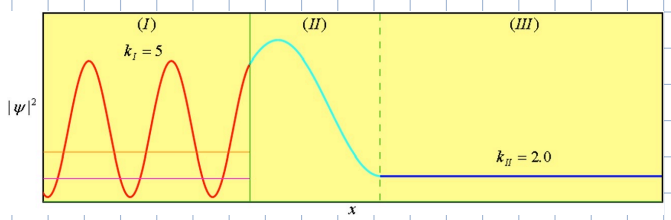
$$C = \frac{4k_1k_2e^{-ik_1a}}{(k_1+k_2)^2e^{-ik_2a} - (k_1-k_2)^2e^{ik_2a}}A$$

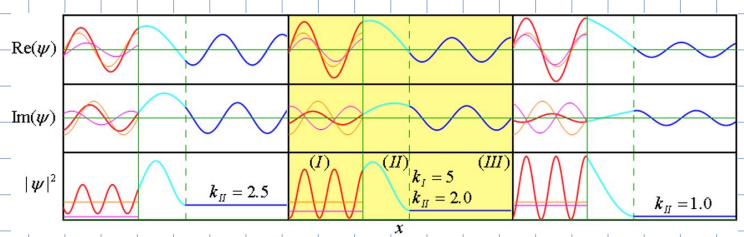
波函数

$$\psi_1 = Ae^{ik_1x} + A'e^{-ik_1x}$$

$$\psi_2 = Be^{ik_2x} + B'e^{-ik_2x}$$

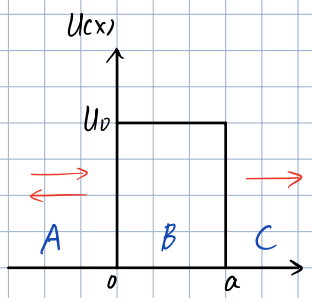
$$\psi_3 = Ce^{ik_1x}.$$





如图所示，A区域中有向右的入射波，向左的反射波.

C区域中有向右的透射波.



入射波

$$\psi_{1\rightarrow} = Ae^{ik_1x}$$

反射波

$$\psi_{1\leftarrow} = A'e^{-ik_1x}$$

透射波

$$\psi_{3\rightarrow} = Ce^{ik_1x}.$$

由概率流密度

$$J=\frac{i\hbar}{2\mu}\left(\psi\frac{d\psi^*}{dx}-\psi^*\frac{d\psi}{dx}\right)$$

入射波

$$J=\frac{i\hbar}{2\mu}\left(\psi_{1\to}\frac{d\psi_{1\to}^*}{dx}-\psi_{1\to}^*\frac{d\psi_{1\to}}{dx}\right)$$

$$=\frac{i\hbar}{2\mu}\left[Ae^{ik_1x}\frac{dA^*e^{-ik_1x}}{dx}-A^*e^{-ik_1x}\frac{d(Ae^{ik_1x})}{dx}\right]$$

$$=\frac{i\hbar}{2\mu}\left[Ae^{ik_1x}\cdot A^*(-ik_1)e^{-ik_1x}-A^*e^{ik_1x}\cdot A(ik_1)e^{ik_1x}\right]$$

$$=\frac{i\hbar}{2\mu}\left(-2|A|^2ik_1\right)=\frac{\hbar k_1}{\mu}|A|^2$$

反射波.

$$J_R=\frac{i\hbar}{2\mu}\left(\psi_{1\leftarrow}\frac{d\psi_{1\leftarrow}^*}{dx}-\psi_{1\leftarrow}^*\frac{d\psi_{1\leftarrow}}{dx}\right)=\frac{\hbar k_1}{\mu}|A'|^2$$

dim 反射系数

$$R\equiv\frac{J_R}{J}=\left|\frac{A'}{A}\right|^2$$

透射波

$$J_D=\frac{i\hbar}{2\mu}\left(\psi_{3\to}\frac{d\psi_{3\to}^*}{dx}-\psi_{3\to}^*\frac{d\psi_{3\to}}{dx}\right)=\frac{\hbar k_1}{\mu}|C|^2$$

dim 透射系数.

$$D\equiv\frac{J_D}{J}=\left|\frac{C}{A}\right|^2$$

概率守恒：

$$R+D=1,\quad J=J_R+J_D.$$

(2) $E < U_0$ 情形

$$\begin{cases} 0 = \dfrac{d^2\psi_2}{dx^2} - k_2^2\psi_2 \\ 0 = \dfrac{d^2\psi_{1,3}}{dx^2} + k_1^2\psi_{1,3} \end{cases}$$

其中，$k_1^2 = \dfrac{2\mu E}{\hbar^2}$，$k_2^2 = \dfrac{2\mu(E-U_0)}{\hbar^2}$

解法与上面相同

波函数

$$\psi_1 = Ae^{ik_1x} + A'e^{-ik_1x}$$

$$\psi_2 = Be^{ik_2x} + B'e^{-ik_2x}$$

$$\psi_3 = Ce^{ik_1x}$$

$$\begin{cases} A' = \dfrac{-i4k_1k_2e^{-ik_1a}}{(k_1-ik_2)^2e^{-k_2a} - (k_1+ik_2)^2e^{k_2a}}A \\ B = \dfrac{+2k_1(k_1+ik_2)e^{-k_2a}}{(k_1-ik_2)^2e^{-k_2a} - (k_1+ik_2)^2e^{k_2a}}A \\ B' = \dfrac{-k_1(k_1+ik_2)e^{k_2a}}{(k_1-ik_2)^2e^{-k_2a} - (k_1+ik_2)^2e^{k_2a}}A \\ C = \dfrac{-i4k_1k_2e^{-k_1a}}{(k_1-ik_2)^2e^{-k_2a} - (k_1+ik_2)^2e^{k_1a}}A \end{cases}$$

反射系数

$$R = \frac{(k_1^2+k_2^2)^2\sh^2k_2a}{(k_1^2+k_2^2)^2\sh^2k_2a + 4k_1^2k_2^2}$$

$$D = \frac{4k_1^2k_2^2}{(k_1^2+k_2^2)^2\sh^2k_2a + 4k_1^2k_2^2}$$

线性谐振子

$$U(x)=\frac{1}{2}\mu\omega^2x^2$$

$$E\psi=-\frac{\hbar^2}{2\mu}\frac{d^2\psi}{dx^2}+U\psi=-\frac{\hbar^2}{2\mu}\frac{d^2\psi}{dx^2}+\frac{1}{2}\mu\omega^2x^2\psi$$

无量纲化：

$$0=\frac{\hbar^2}{2\mu}\frac{d^2\psi}{dx^2}+\left(E-\frac{\mu\omega^2}{2}x^2\right)\psi=\frac{\hbar^2}{2\mu}\alpha^2\frac{d^2\psi}{d\xi^2}+\left(E-\frac{\mu\omega^2}{2\alpha^2}\xi^2\right)\psi$$

$$\Longleftrightarrow\quad \frac{d^2\psi}{d\xi^2}+\frac{2\mu}{\hbar^2}\left(E-\frac{\mu\omega^2}{2\alpha^2}\xi^2\right)\psi=0$$

$$\alpha=\sqrt{\frac{\mu\omega}{\hbar}}\qquad \xi=\alpha x$$

令 $\lambda=\frac{2\mu E}{\hbar^2\alpha^2}=\frac{2\mu E}{\hbar^2}\frac{\hbar}{\mu\omega}=\frac{2E}{\hbar\omega}$

化为：

$$\frac{d^2\psi}{d\xi^2}+(\lambda-\xi^2)\psi=0$$

$$\xi=\alpha x,\quad \alpha=\sqrt{\frac{\mu\omega}{\hbar}},\quad \lambda=\frac{2E}{\hbar\omega}$$

$$\psi(\pm\infty)=0$$

边界条件 $\xi\to\pm\infty$ 时

$$0=\frac{d^2\psi}{d\xi^2}+(\lambda-\xi^2)\psi=\frac{d^2\psi}{d\xi^2}-\xi^2\psi$$

$$\lim_{\xi\to\infty}(\lambda-\xi^2)=-\xi^2$$

$$\psi(\xi)=Ae^{\frac{\xi^2}{2}}+Be^{-\frac{\xi^2}{2}},$$

$$\psi(\pm\infty)=A\infty+B\cdot 0\quad\Longrightarrow\quad A=0,\quad \psi=Be^{-\frac{\xi^2}{2}}$$

我们对波函数前的常系数不感兴趣，因为归一化时能直接并入归一化因子中.

于是波函数在无穷远处.

$$\psi \propto e^{-\frac{\xi^2}{2}}$$

非无穷远处，假设

$$\psi(\xi) = e^{-\frac{\xi^2}{2}} H(\xi) \quad ?$$

$$\psi'_\xi = -\xi e^{-\frac{\xi^2}{2}} H + e^{-\frac{\xi^2}{2}} H'_\xi$$

$$\psi''_{\xi\xi} = -e^{-\frac{\xi^2}{2}} H + \xi^2 e^{-\frac{\xi^2}{2}} H - 2\xi e^{-\frac{\xi^2}{2}} H'_\xi + e^{-\frac{\xi^2}{2}} H''_\xi$$

代入方程中，从而有 $H(\xi)$ 的 eq：

$$\psi''_{\xi\xi} + (\lambda - \xi^2)\psi = 0$$

$$\underline{H''_{\xi\xi} - 2\xi H'_\xi + (\lambda - 1) H = 0}$$

考虑，级数解：

$$H(\xi) = \sum_{v=0}^{\infty} a_v \xi^v$$

$$H'_\xi = \sum_{v=0}^{\infty} v a_v \xi^{v-1}$$

$$H''_{\xi\xi} = \sum_{v=2}^{\infty} v(v-1) a_v \xi^{v-2} = \sum_{v=0}^{\infty} (v+1)(v+2) a_{v+2} \xi^v$$

代入，有：

$$0 = H''_{\xi\xi} - 2\xi H'_\xi + (\lambda - 1) H$$

$$= \sum_{v=0}^{\infty} (v+1)(v+1) a_{v+2} \xi^v - 2\xi \sum_{v=0}^{\infty} v a_v \xi^{v-1} + (\lambda - 1) \sum_{v=0}^{\infty} a_v \xi^v$$

$$= \sum_{v=0}^{\infty} \xi^v \left[a_{v+2}(v+1)(v+2) - 2v a_v + (\lambda-1) a_v \right] \equiv 0$$

$$\Longrightarrow$$

$$a_{v+2}(v+1)(v+2) + (\lambda - 2v - 1) a_v = 0.$$

$$\frac{a_{v+2}}{a_v} = \frac{2v-\lambda+1}{(v+2)(v+1)}$$

我们已知：波函数不应发散

考虑，

$$\lim_{v\to\infty} \frac{a_{v+2}}{a_v} = \lim_{v\to\infty} \frac{2v-\lambda+1}{(v+2)(v+1)} \longrightarrow \frac{2}{v}$$

再考虑函数.

$$e^{\xi^2} = \sum_{v=0}^{\infty} b_v \xi^v \quad \Longrightarrow \quad b_v = \begin{cases} \left[(\frac{v}{2})! \right]^{-1} & v \in \text{even} \\ 0 & v \in \text{odd} \end{cases}$$

$$\frac{b_{v+2}}{b_v} = \frac{(\frac{v}{2})!}{(\frac{v}{2}+1)!} = \frac{1}{\frac{v}{2}+1}$$

$$\lim_{v\to\infty} \frac{b_{v+2}}{b_v} = \frac{2}{v}$$

我们看到，当 $v \to \infty$ 时，2级数 $\sum_{v=0}^{\infty} a_v \xi^v$ 与 $\sum_{v=0}^{\infty} b_v \xi^v$ 应有相同的行径.

$$\lim_{v\to\infty} H \longrightarrow e^{\xi^2}$$

$$\lim_{v\to\infty} \psi = \lim_{v\to\infty} e^{-\frac{\xi^2}{2}} H \longrightarrow \lim_{v\to\infty} e^{-\frac{\xi^2}{2}} e^{\xi^2} = \lim_{v\to\infty} e^{\frac{\xi^2}{2}} \longrightarrow \infty.$$

而

$$\lim_{v\to\infty} \psi \neq \infty.$$

∴ 幂级数 H 应在某处截断，即 ν 取某值时

$$2\nu-\lambda+1=0.$$

$$\lambda=2n+1,\quad n=1,2,\cdots.$$

同时注意，由于上述关系仅约束了 a_ν 与 $a_{\nu+2}$，所以只能确保级数的偶数项与奇数项二者之一被截断. ∴ 若偶数项被截断，奇数项要恒为0. 反之亦然.

$$\begin{cases} a_0=0 & n\in odd \\ a_1=0 & n\in even \end{cases}$$

综上，$\lambda=2n+1,\ n=0,1,2,\cdots$；$\lambda=\frac{2E}{\hbar\omega}$

$$E=\left(n+\frac{1}{2}\right)\hbar\omega$$

$$\varphi_n(x)=N_n e^{-\frac{\alpha^2x^2}{2}}H(\alpha x).$$

N_n：归一化因子　　$H(\alpha x)$：Hermite polynomial $H_n(\xi)$

$$N_n=\left(\frac{\alpha}{\sqrt{\pi}2^n n!}\right)^{\frac{1}{2}}$$

$$H_n(\xi)=\sum_{\nu=0}^{\infty}a_\nu\xi^\nu$$

$$\begin{cases} a_0=0 & n\in odd \\ a_1=0 & n\in even \end{cases}$$

$$\frac{a_{\nu+2}}{a_\nu}=\frac{2(\nu-n)}{(\nu+1)(\nu+2)}$$

能谱

$$E_n = \left(n+\frac{1}{2}\right)\hbar\omega \qquad n \in \mathbb{Z}.$$

零点能

$$E_0 = \frac{1}{2}\hbar\omega.$$

相等的能级间距.

$$\Delta E = E_{n+1} - E_n = \hbar\omega$$

宇称

定理：若 $U(-x)=U(x)$，且 $\psi(x)$ 为定态薛定谔方程的解，那么 $\psi(-x)$ 也是属于同一个本征值 E 的解。

证明：

$$E\psi(x)=-\frac{\hbar^2}{2\mu}\frac{d^2\psi(x)}{dx^2}+U(x)\psi(x),$$

$$U(x)=U(-x)$$

$$E\psi(-x)=-\frac{\hbar^2}{2\mu}\frac{d^2\psi(-x)}{d(-x)^2}+U(x)\psi(-x)=-\frac{\hbar^2}{2\mu}\frac{d^2\psi(-x)}{dx^2}+U(x)\psi(-x).$$

定理：若 $U(x)=U(-x)$，则定态薛定谔方程的每个解都有确定的宇称。

（奇偶性：$\psi(-x)=\psi(x)$ 或 $\psi(-x)=-\psi(x)$）。

证明：设 $\psi(x)$ 为方程的解，则 $\psi(-x)$ 也是方程的解。

构造

$$\psi_S(x)=\psi(x)+\psi(-x)$$

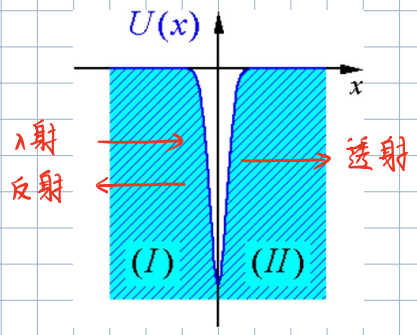
$$\psi_A(x)=\psi(x)-\psi(-x).$$

显然，

$\psi_S(-x)=\psi_S(x)$ 为偶宇称

$\psi_A(-x)=-\psi_A(x)$ 为奇宇称。

δ势阱 / 势垒.



一维δ势阱

$$U(x) = -r\delta(x) \qquad r>0$$

$$0 = \frac{\hbar^2}{2\mu}\frac{\partial^2\psi}{\partial x^2} - U\psi + E\psi$$

① $x\neq 0$, $U=0$

$$0 = \frac{\hbar}{2\mu}\frac{\partial^2\psi}{\partial x^2} + E\psi$$

② $x=0$ $\quad U=-r\delta(x)$.

$$0 = \frac{\hbar^2}{2\mu}\frac{\partial^2\psi}{\partial x^2} + E\psi + r\delta(x).$$

$$0 = \frac{\hbar^2}{2\mu}\int_{-\varepsilon}^{\varepsilon}\frac{d^2\psi}{dx^2}\,d\xi + r\int_{-\varepsilon}^{\varepsilon}\delta(x)\psi\,d\xi + E\int_{-\varepsilon}^{\varepsilon}\psi\,d\xi$$

$$= \frac{\hbar^2}{2\mu}\psi'\Big|_{-\varepsilon}^{\varepsilon} + r\psi(0) + \underbrace{E\,2\varepsilon\,\psi(x')}_{\to 0\ (\varepsilon\to 0)} \qquad (-\varepsilon<x'<\varepsilon)$$

$$= \frac{\hbar^2}{2\mu}\psi'\Big|_{0^-}^{0^+} + r\psi(0)$$

$\varepsilon\to 0$, 有:

$$\psi_2'(0) - \psi_1'(0) = -\frac{2\mu r}{\hbar^2}\psi(0).$$

微分方程的边界问题.

① $E<0$.

$$\frac{d^2\psi_{1,2}}{dx^2}+\frac{2\mu E}{\hbar^2}\psi_{1,2}=\frac{d^2\psi_{1,2}}{dx^2}-k^2\psi_{1,2}=0.$$

$$\psi_1'(0)-\psi_2'(0)=-\frac{2\mu\gamma}{\hbar^2}\psi_{1,2}(0)=-g\psi_{1,2}(0)$$

$$\psi_1(0)=\psi_2(0)$$

求解:

$$\psi_1=Ae^{kx}+A'e^{-kx}$$

$$\psi_2=Be^{kx}+B'e^{-kx}$$

$$\psi(\pm\infty)\neq\infty \Longrightarrow A'=0 \quad B=0.$$

$$\psi_1=Ae^{kx}$$

$$\psi_2=B'e^{-kx}$$

$$\psi_1(0)=\psi_2(0) \Longrightarrow A=B'=C$$

$$\psi_2'(0)-\psi_1'(0)=-2kC=-gC$$

$$E=-\frac{\mu\gamma^2}{2\hbar^2}$$

② $E=0$

$$\psi_1=C_1x \quad , \quad \psi_2=C_2x.$$

$$\psi_2'(0)-\psi_1'(0)=C_1-C_2=0 \Longrightarrow C_2=C_1=C$$

$$\psi=Cx.$$

$$\psi(\pm\infty) \neq 0 \quad \Longrightarrow \quad C=0 \quad \psi=0.$$

无意义.

③ $E>0$.

$$\begin{cases} \dfrac{d^2\psi_{1,2}}{dx^2}+\dfrac{2\mu E}{\hbar^2}\psi_{1,2}=\dfrac{d^2\psi_{1,2}}{dx^2}+k^2\psi_{1,2}=0 \\ \psi_1'(0)-\psi_2'(0)=-\dfrac{2\mu\gamma}{\hbar^2}\psi_{1,2}(0)=-g\psi_{1,2}(0) \\ \psi_1(0)=\psi_2(0). \end{cases}$$

考虑,

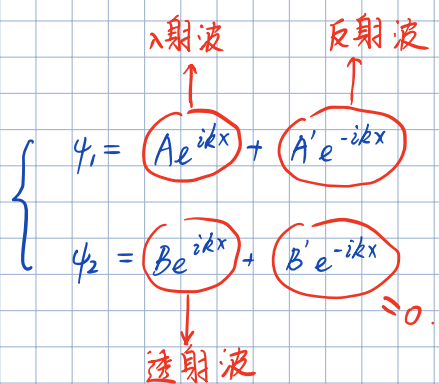
$$\begin{cases} \psi_1 = Ae^{ikx}+A'e^{-ikx} \\ \psi_2 = Be^{ikx}+B'e^{-ikx} \end{cases}$$

则:

$$\psi_1(0)=\psi_2(0) \Longrightarrow A+A'=B+B'$$

$$\psi_2'(0)-\psi_1'(0)=ik(B-B'-A+A')=-\frac{2\mu\gamma}{\hbar^2}(A+A')$$

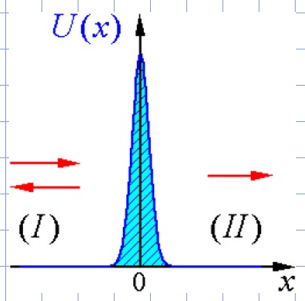
考虑波函数的物理意义:



在Ⅱ区域没有由右向左的波. $B'=0$.

联立可求解, 后续省略.

例：一维 δ 势垒.



$$U(x) = \gamma\delta(x), \quad \gamma > 0$$

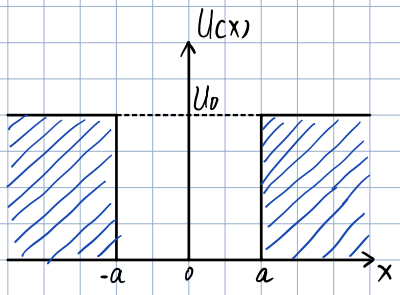
定态薛定谔方程

$$0 = \frac{\hbar^2}{2\mu}\frac{d^2\psi_{1,2}}{dx^2} + E\psi_{1,2} \qquad x \neq 0$$

$$\psi_2'(0) - \psi_1'(0) = \frac{2\mu\gamma}{\hbar^2}\psi_{1,2}(0).$$

求解与上面相同略去

有限深势阱



$$U(x)=\begin{cases}0 & |x|<a\\ U_0 & |x|>a\end{cases}.$$

定态薛定谔方程

1. $E<U_0$

$$\begin{cases}-\frac{\hbar^2}{2\mu}\frac{d^2\psi}{dx^2}=E\psi, & |x|<a\\ -\frac{\hbar^2}{2\mu}\frac{d^2\psi}{dx^2}+U_0\psi=E\psi, & |x|\geq a\end{cases}$$

$$\Rightarrow\begin{cases}\frac{d^2\psi}{dx^2}+k^2\psi=0, \quad k=\sqrt{\frac{2\mu E}{\hbar^2}} & |x|<a\\ \frac{d^2\psi}{dx^2}-k'^2\psi=0 \quad k'=\sqrt{\frac{2\mu(U_0-E)}{\hbar^2}} & |x|\geq a\end{cases}$$

$|x|<a$.

$$\psi(x)=\begin{cases}Ae^{ikx}\\ Be^{-ikx}\end{cases}$$

$|x|\geq a$

$$\psi(x)=\begin{cases}Ce^{k'x}\\ De^{-k'x}\end{cases}$$

波函数连续性条件

$$\left.\begin{aligned}(\psi_1)_{x=a} &= (\psi_2)_{x=a}\\ (\frac{d\psi_1}{dx})_{x=a} &= (\frac{d\psi_2}{dx})_{x=a}\end{aligned}\right\} \Rightarrow (\frac{1}{\psi_1}\frac{d\psi_1}{dx})_{x=a} = (\frac{1}{\psi_2}\frac{d\psi_2}{dx})_{x=a}$$

$$(\frac{d\ln\psi_1}{dx})_{x=a} = (\frac{d\ln\psi_2}{dx})_{x=a}$$

偶宇称时，

$$\left.\begin{aligned}[\frac{d\ln(A\cos kx)}{dx}]_{x=a} &= (\frac{-k\sin kx}{\cos kx})_{x=a} = -k\tan(ka)\\ [\frac{d\ln Ce^{-k'x}}{dx}]_{x=a} &= [\frac{d(-k'x)}{dx}]_{x=a} = -k'\end{aligned}\right\} \Rightarrow k\tan(ka)=k'$$

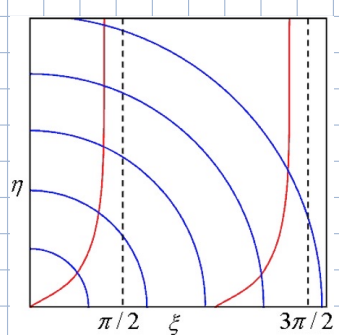
$x=-a$ 处同理.

令 $\xi = ka \quad \eta = k'a \Rightarrow \xi\tan\xi = \eta$

$$k^2 = \frac{2\mu E}{\hbar^2}$$

$$k'^2 = \frac{2\mu(U_0-E)}{\hbar^2}$$

$$k^2+k'^2 = \frac{2\mu U_0}{\hbar^2} \Rightarrow \xi^2+\eta^2 = \frac{2\mu U_0 a^2}{\hbar^2}$$



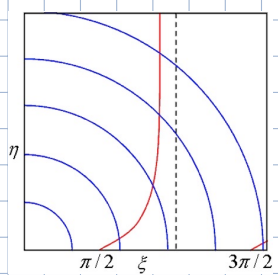
奇宇称闭态

$$\left[\frac{d\ln(A\sin kx)}{dx}\right]_{x=a} = \left(\frac{k\cos kx}{\sin kx}\right)_{x=a} = k\cot(ka)$$

$$\left[\frac{d\ln Ce^{-k'x}}{dx}\right]_{x=a} = \left[\frac{d(-k'x)}{dx}\right]_{x=a} = -k'$$

$$\Big\} \; -k\cot(ka) = k'$$

令 $\xi = ka \quad \eta = k'a \quad \Rightarrow \quad -\xi\cot\xi = \eta$



$$k^2 + k'^2 = \frac{2\mu U_0}{\hbar^2} \Longrightarrow \xi^2 + \eta^2 = \frac{2\mu U_0 a^2}{\hbar^2}$$

对任意 U_0a^2，一定有偶宇称闭态；

仅当 $U_0a^2 \geq \frac{\pi^2\hbar^2}{8\mu}$，才有奇宇称态.

当 $U_0 \to \infty$，退化为一维无限深势阱.

$$\frac{n\pi}{2} = \xi = ka = \sqrt{\frac{2\mu E}{\hbar^2}}a \quad \Rightarrow \quad E = \frac{\pi^2\hbar^2n^2}{8\mu a^2}, \quad n = 1, 2, \cdots$$

2、$E > U_0$

$$\begin{cases} \frac{d^2\psi}{dx^2} + k^2\psi = 0 & k = \sqrt{\frac{2\mu E}{\hbar^2}} & |x| < a \\ \frac{d^2\psi}{dx^2} + k'^2\psi = 0 & k' = \sqrt{\frac{2\mu(E-U_0)}{\hbar^2}} & |x| \geq a. \end{cases}$$

$$\psi_1(x) = Ae^{ik'x} + A'e^{-ik'x}$$

$$\psi_2(x) = Be^{ikx} + B'e^{-ikx}$$

$$\psi_3(x) = Ce^{ik'x}$$

波函数连续性条件：

$$\begin{cases} Ae^{-ik'a} + A'e^{ik'a} - Be^{-ika} - B'e^{ika} = 0 \\ Ak'e^{-ika} - A'k'e^{ik'a} - Bke^{-ika} + B'e^{ika} = 0 \\ Be^{ika} + B'e^{-ika} - Ce^{ik'a} = 0 \\ Bke^{ika} - B'ke^{-ika} - Ck'e^{ik'a} = 0 \end{cases}$$

$$\Rightarrow \quad C = -\frac{4kk'e^{i2a(k-k')}}{e^{i4ak}(k-k')^2 - (k+k')^2} A$$

$$D = \frac{J_D}{J} = \left|\frac{C}{A}\right|^2 = \left|-\frac{4kk'e^{i2a(k-k')}}{e^{i4ak}(k-k')^2 - (k+k')^2}\right|^2$$

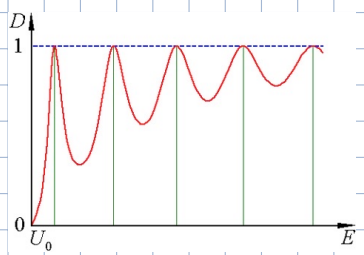
$$= \left[1 + \frac{U_0^2 \sin^2(2ka)}{4E(E-U_0)}\right]^{-1}$$

- $E > U_0$ 时，D 可小于 1. 粒子可以反射.
- 可取合适 E 使 $D=1$，称共振透射. 此时 E 称共振能级.

$$\sin^2(2ka) = 0 \quad \Rightarrow \quad k = \frac{n\pi}{2a}$$

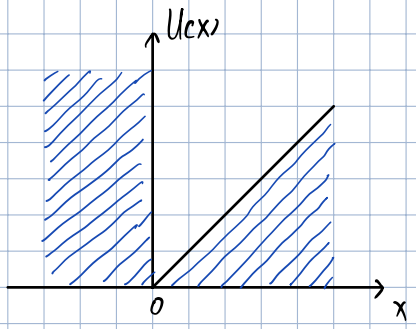
$$\lambda = 2a\frac{2}{n}$$

$$E_n = \frac{n^2\pi^2\hbar^2}{8\mu a^2}$$



线性势阱

$$U(x)=\begin{cases}\infty & x<0\\ Fx & x\geq 0\end{cases}\qquad (F>0)$$



定态薛定谔方程.

$$-\frac{\hbar^2}{2\mu}\frac{d^2}{dx^2}\psi+Fx\psi=E\psi$$

化为:

$$\frac{d^2}{dx^2}\psi-\frac{2\mu}{\hbar^2}(Fx-E)\psi=0.$$

换元, 令 $y=ax+b$, $x=\frac{y-b}{a}$, 有:

$$\frac{d}{dx}=a\frac{d}{dy}\qquad \frac{d^2}{dx^2}=a^2\frac{d^2}{dy^2}$$

代入,有:

$$a^2\frac{d^2}{dy^2}\psi-\frac{2\mu}{\hbar^2}\left[F\left(\frac{y}{a}-b\right)-E\right]\psi=0.$$

$$\frac{d^2}{dy^2}\psi-\frac{2\mu}{\hbar^2a^2}\left[F\frac{y}{a}-(Fb+E)\right]\psi=0.$$

$\Longrightarrow$

$$\frac{2\mu}{a^2\hbar^2}\frac{F}{a}=1,$$

$$a=\left(\frac{2\mu F}{\hbar^2}\right)^{\frac{1}{3}},\quad b=-\frac{E}{F}$$

$$y=\left(\frac{2\mu F}{\hbar^2}\right)^{\frac{1}{3}}\left(x-\frac{E}{F}\right)$$

方程化为:

$$\begin{cases}\psi=0, & y<-\left(\frac{2\mu E^3}{\hbar^2F^2}\right)^{\frac{1}{3}}\\ \frac{d^2}{dy^2}\psi-y\psi=0, & y=\left(\frac{2\mu F}{\hbar^2}\right)^{\frac{1}{3}}\left(x-\frac{E}{F}\right),\quad y\geq-\left(\frac{2\mu E^3}{\hbar^2F^2}\right)^{\frac{1}{3}}\end{cases}$$

边界条件：

$$(\psi)_{x=0} = (\psi)_{x=\infty} = 0 \implies (\psi)_{y=-(\frac{2\mu E^3}{\hbar^2 F^2})^{\frac{1}{3}}} = (\psi)_{y=\infty} = 0$$

求解：

① $y>0$. $E<Fx$（总能量小于势能）称经典禁区.

令 $\xi = \frac{2}{3}y^{\frac{3}{2}}$ $\quad u = \frac{1}{\sqrt{y}}\psi$.

微分方程化为 Bessel 方程.

$$\frac{d^2u}{d\xi^2} + \frac{1}{\xi}\frac{du}{d\xi} - [1 + \frac{(\frac{1}{3})^2}{\xi^2}]u = 0.$$

$$\implies \quad u = AK_{\frac{1}{3}}(\xi) + A'I_{\frac{1}{3}}(\xi)$$

代入边界条件：

$$(\psi)_{y=\infty} = 0$$

$$A' = 0,$$

$$\psi_1 = \sqrt{y}u = A\sqrt{y}\,K_{\frac{1}{3}}(\frac{2}{3}y^{\frac{3}{2}})$$

② 经典允许区，$-(\frac{2\mu E^3}{\hbar^2 F^2})^{\frac{1}{3}} < y < 0$.

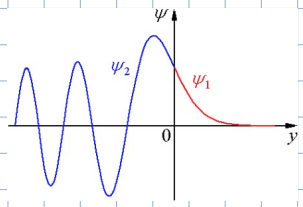
令 $\xi = \frac{2}{3}|y|^{\frac{3}{2}}$ $\quad u = \frac{1}{\sqrt{y}}\psi$. 化为 Bessel eq

$$\frac{d^2u}{d\xi^2} + \frac{1}{\xi}\frac{du}{d\xi} + [1 - \frac{(\frac{1}{3})^2}{\xi^2}]u = 0. \implies u = BJ_{\frac{1}{3}}(\xi) + B'J_{-\frac{1}{3}}(\xi)$$

波函数连续性：

$$(\psi_1)_{y=0} = (\psi_2)_{y=0} \implies B = B' = \frac{\pi}{\sqrt{3}}A.$$

$$\psi_2 = \sqrt{|y|}u = \frac{\pi}{\sqrt{3}}A\sqrt{|y|}[J_{\frac{1}{3}}(\frac{2}{3}(|y|)^{\frac{3}{2}} + J_{-\frac{1}{3}}(\frac{2}{3}|y|^{\frac{3}{2}})]$$

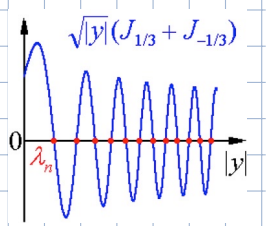


• 束缚态能级.

$$(\psi)_{y=-(\frac{2\mu E}{\hbar^2F^2})^{\frac{1}{3}}}=0 \quad \Longrightarrow \quad J_{\frac{1}{3}}(\tfrac{2}{3}|y|^{\frac{3}{2}})+J_{-\frac{1}{3}}(\tfrac{2}{3}|y|^{\frac{3}{2}})=0.$$

$$|y|=\lambda_n, \quad n=1,2,3.$$

$$E_n=(\frac{\hbar^2F^2}{2\mu})^{\frac{1}{3}}\lambda_n.$$



• 以一维重力场为例.

$$U(x)=\mu g x.$$

$$E_n=(\frac{\hbar^2\mu g^2}{2})^{\frac{1}{3}}\lambda_n.$$

经典禁典分界.

$$x_0=\frac{E_n}{\mu g}=-(\frac{2E_n^3}{\mu g^2\hbar^2})^{\frac{1}{3}}.$$

